Proximal Policy Optimization: A Comprehensive Mathematical Analysis

Formal Treatment with Complete Proofs

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Abstract

This document provides a rigorous mathematical treatment of the Proximal Policy Optimization (PPO) algorithm, including complete proofs of convergence properties, sample efficiency analysis, loss function derivation, network architecture considerations, and backpropagation through the policy network. We establish theoretical guarantees for PPO's monotonic improvement property and provide detailed analysis of its computational and statistical properties.

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1 Introduction

Proximal Policy Optimization (PPO) represents a significant advancement in policy gradient methods for reinforcement learning. Developed by Schulman et al. (2017), PPO addresses fundamental challenges in policy optimization: maintaining stable learning while maximizing sample efficiency.

1.1 Motivation and Context

Traditional policy gradient methods suffer from:

1. High sensitivity to step size selection
2. Poor sample efficiency requiring extensive interaction
3. Lack of monotonic improvement guarantees
4. Tendency to take destructively large policy updates

PPO introduces a novel clipped objective that provides a principled approach to constraining policy updates while maintaining computational efficiency.

2 Mathematical Foundations

2.1 Markov Decision Process Framework

Definition 2.1 (Markov Decision Process)**.** A Markov Decision Process (MDP) is a tuple $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, r, \rho_0, \gamma)$ where:

- $\mathcal{S}$ is the state space
- $\mathcal{A}$ is the action space
- $P : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \to [0, 1]$ is the transition probability
- $r : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$ is the reward function
- $\rho_0 : \mathcal{S} \to [0, 1]$ is the initial state distribution
- $\gamma \in [0, 1)$ is the discount factor

Definition 2.2 (Stochastic Policy)**.** A stochastic policy $\pi_\theta : \mathcal{S} \times \mathcal{A} \to [0, 1]$ parameterized by $\theta \in \mathbb{R}^d$ specifies the probability of selecting action a in state s:

$$\pi_\theta(a|s) = \Pr(A_t = a|S_t = s; \theta)$$

2.2 Value Functions and Performance Measures

Definition 2.3 (State Value Function)**.** The state value function under policy π is:

$$V^\pi(s) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t r(S_t, A_t) \,\middle|\, S_0 = s \right]$$

Definition 2.4 (State-Action Value Function)**.** The state-action value function (Q-function) under policy π is:

$$Q^\pi(s, a) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t r(S_t, A_t) \,\middle|\, S_0 = s, A_0 = a \right]$$

Definition 2.5 (Advantage Function)**.** The advantage function measures the relative value of taking action a in state s:

$$A^\pi(s, a) = Q^\pi(s, a) - V^\pi(s)$$

Definition 2.6 (Policy Performance)**.** The expected return of policy π is:

$$J(\pi) = \mathbb{E}_{s_0 \sim \rho_0} \left[V^\pi(s_0) \right] = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t r(S_t, A_t) \right]$$

2.3 Policy Improvement Theory

The foundation of PPO rests on the Policy Improvement Theorem, which we state and prove rigorously.

Theorem 2.7 (Policy Improvement Theorem)**.** Let π and $\tilde{\pi}$ be two policies. Then:

$$J(\tilde{\pi}) = J(\pi) + \mathbb{E}_{\tau \sim \tilde{\pi}} \left[\sum_{t=0}^{\infty} \gamma^t A^\pi(S_t, A_t) \right]$$

where τ denotes a trajectory $(S_0, A_0, S_1, A_1, \ldots)$.

Proof. We begin by expanding the expected return under $\tilde{\pi}$:

$$\begin{aligned} J(\tilde{\pi}) &= \mathbb{E}_{\tilde{\pi}} \left[\sum_{t=0}^{\infty} \gamma^t r(S_t, A_t) \right] \\ &= \mathbb{E}_{\tilde{\pi}} \left[r(S_0, A_0) + \gamma \sum_{t=1}^{\infty} \gamma^{t-1} r(S_t, A_t) \right] \end{aligned}$$

Now we use the recursive structure of the value function. For any policy π':

$$\begin{aligned} V^{\pi'}(s) &= \mathbb{E}_{\pi'} \left[r(s, a) + \gamma V^{\pi'}(S_{t+1}) | S_t = s \right] \\ &= \sum_a \pi'(a|s) \left[r(s, a) + \gamma \sum_{s'} P(s'|s, a) V^{\pi'}(s') \right] \end{aligned}$$

Consider the difference:

$$\begin{aligned} J(\tilde{\pi}) - J(\pi) &= V^{\tilde{\pi}}(s_0) - V^{\pi}(s_0) \\ &= \mathbb{E}_{\tilde{\pi}}[r(S_0, A_0) + \gamma V^{\tilde{\pi}}(S_1)] - V^{\pi}(s_0) \end{aligned}$$

Add and subtract $\gamma V^{\pi}(S_1)$:

$$\begin{aligned} &= \mathbb{E}_{\tilde{\pi}}[r(S_0, A_0) + \gamma V^{\pi}(S_1) - V^{\pi}(S_0)] + \gamma \mathbb{E}_{\tilde{\pi}}[V^{\tilde{\pi}}(S_1) - V^{\pi}(S_1)] \\ &= \mathbb{E}_{\tilde{\pi}}[A^{\pi}(S_0, A_0)] + \gamma \mathbb{E}_{\tilde{\pi}}[V^{\tilde{\pi}}(S_1) - V^{\pi}(S_1)] \end{aligned}$$

Applying this recursively to $V^{\tilde{\pi}}(S_1) - V^{\pi}(S_1)$:

$$= \mathbb{E}_{\tilde{\pi}}[A^{\pi}(S_0, A_0)] + \gamma \mathbb{E}_{\tilde{\pi}}[A^{\pi}(S_1, A_1)] + \gamma^2 \mathbb{E}_{\tilde{\pi}}[V^{\tilde{\pi}}(S_2) - V^{\pi}(S_2)]$$

Continuing this expansion indefinitely:

$$J(\tilde{\pi}) - J(\pi) = \sum_{t=0}^{\infty} \gamma^t \mathbb{E}_{\tilde{\pi}}[A^{\pi}(S_t, A_t)]$$

Therefore:

$$J(\tilde{\pi}) = J(\pi) + \mathbb{E}_{\tau \sim \tilde{\pi}} \left[\sum_{t=0}^{\infty} \gamma^t A^{\pi}(S_t, A_t) \right]$$

□

Corollary 2.8 (Greedy Policy Improvement)**.** If $\tilde{\pi}(s) \in \operatorname{argmax}_a Q^{\pi}(s, a)$ for all states s, then $J(\tilde{\pi}) \geq J(\pi)$.

Proof. Since $\tilde{\pi}$ selects actions that maximize $Q^{\pi}(s, a)$, we have:

$$Q^{\pi}(s, \tilde{\pi}(s)) \geq Q^{\pi}(s, a) \quad \forall a \in \mathcal{A}$$

Therefore:

$$A^{\pi}(s, \tilde{\pi}(s)) = Q^{\pi}(s, \tilde{\pi}(s)) - V^{\pi}(s) \geq 0$$

By Theorem 2.7:

$$J(\tilde{\pi}) = J(\pi) + \mathbb{E}_{\tau \sim \tilde{\pi}} \left[\sum_{t=0}^{\infty} \gamma^t A^{\pi}(S_t, A_t) \right] \geq J(\pi)$$

□

3 Trust Region Policy Optimization Background

PPO builds upon Trust Region Policy Optimization (TRPO). We present the key results from TRPO that motivate PPO's design.

3.1 Local Approximation of Policy Performance

Definition 3.1 (Discounted State Visitation Distribution)**.** The discounted state visitation distribution under policy π is:

$$d^{\pi}(s) = (1-\gamma)\sum_{t=0}^{\infty}\gamma^t \Pr(S_t = s|\pi)$$

where the probability is taken over trajectories generated by π starting from $s_0 \sim \rho_0$.

Lemma 3.2 (Alternative Performance Difference Formula)**.** For any two policies π and $\tilde{\pi}$:

$$J(\tilde{\pi}) = J(\pi) + \frac{1}{1-\gamma}\mathbb{E}_{s\sim d^{\tilde{\pi}}, a\sim\tilde{\pi}(\cdot|s)}[A^{\pi}(s,a)]$$

Proof. From Theorem 2.7:

$$J(\tilde{\pi}) - J(\pi) = \mathbb{E}_{\tau\sim\tilde{\pi}}\left[\sum_{t=0}^{\infty}\gamma^t A^{\pi}(S_t, A_t)\right]$$

Rearranging the expectation over trajectories into expectations over the state visitation distribution:

$$\begin{aligned}\mathbb{E}_{\tau\sim\tilde{\pi}}\left[\sum_{t=0}^{\infty}\gamma^t A^{\pi}(S_t, A_t)\right] &= \sum_{t=0}^{\infty}\gamma^t \mathbb{E}_{S_t, A_t\sim\tilde{\pi}}[A^{\pi}(S_t, A_t)] \\ &= \sum_{t=0}^{\infty}\gamma^t \sum_s \Pr(S_t = s|\tilde{\pi})\sum_a \tilde{\pi}(a|s)A^{\pi}(s,a)\end{aligned}$$

Define the unnormalized state visitation:

$$\rho^{\tilde{\pi}}(s) = \sum_{t=0}^{\infty}\gamma^t \Pr(S_t = s|\tilde{\pi})$$

Then:

$$J(\tilde{\pi}) - J(\pi) = \sum_s \rho^{\tilde{\pi}}(s)\sum_a \tilde{\pi}(a|s)A^{\pi}(s,a)$$

Note that $\sum_s \rho^{\tilde{\pi}}(s) = 1/(1-\gamma)$ (sum of geometric series over discount factors). Therefore:

$$d^{\tilde{\pi}}(s) = (1-\gamma)\rho^{\tilde{\pi}}(s)$$

Substituting:

$$J(\tilde{\pi}) - J(\pi) = \frac{1}{1-\gamma}\sum_s d^{\tilde{\pi}}(s)\sum_a \tilde{\pi}(a|s)A^{\pi}(s,a)$$

Which gives the desired result. □

3.2 Local Policy Approximation

The challenge with Lemma 3.2 is that it requires knowing $d^{\tilde{\pi}}$, which depends on the new policy $\tilde{\pi}$ we're trying to optimize. TRPO introduces a local approximation.

Definition 3.3 (Surrogate Objective)**.** The surrogate objective function is:

$$L_\pi(\tilde{\pi}) = J(\pi) + \frac{1}{1-\gamma}\mathbb{E}_{s\sim d^\pi, a\sim\tilde{\pi}(\cdot|s)}[A^\pi(s,a)]$$

Note that L_π uses d^π (old policy's state distribution) instead of $d^{\tilde{\pi}}$.

Lemma 3.4 (First-Order Match)**.** The surrogate objective L_π matches J to first order at π:

$$\begin{aligned} L_\pi(\pi) &= J(\pi) \\ \nabla_\theta L_\pi(\pi_\theta)\big|_{\theta=\theta_{\text{old}}} &= \nabla_\theta J(\pi_\theta)\big|_{\theta=\theta_{\text{old}}} \end{aligned}$$

Proof. For the first equality:

$$\begin{aligned} L_\pi(\pi) &= J(\pi) + \frac{1}{1-\gamma}\mathbb{E}_{s\sim d^\pi, a\sim\pi(\cdot|s)}[A^\pi(s,a)] \\ &= J(\pi) + \frac{1}{1-\gamma}\sum_s d^\pi(s)\sum_a \pi(a|s)A^\pi(s,a) \end{aligned}$$

By definition of the value function:

$$V^\pi(s) = \sum_a \pi(a|s)Q^\pi(s,a)$$

Therefore:

$$\sum_a \pi(a|s)A^\pi(s,a) = \sum_a \pi(a|s)[Q^\pi(s,a) - V^\pi(s)] = V^\pi(s) - V^\pi(s) = 0$$

Thus $L_\pi(\pi) = J(\pi)$.
For the gradient equality, we use the policy gradient theorem. The gradient of J is:

$$\nabla_\theta J(\pi_\theta) = \frac{1}{1-\gamma}\mathbb{E}_{s\sim d^{\pi_\theta}, a\sim\pi_\theta(\cdot|s)}[\nabla_\theta \log \pi_\theta(a|s)A^{\pi_\theta}(s,a)]$$

At $\theta = \theta_{\text{old}}$, we have $\pi_\theta = \pi$ and:

$$\begin{aligned} \nabla_\theta L_\pi(\pi_\theta)\big|_{\theta=\theta_{\text{old}}} &= \nabla_\theta\left[\frac{1}{1-\gamma}\mathbb{E}_{s\sim d^\pi, a\sim\pi_\theta(\cdot|s)}[A^\pi(s,a)]\right]_{\theta=\theta_{\text{old}}} \\ &= \frac{1}{1-\gamma}\mathbb{E}_{s\sim d^\pi}\left[\sum_a \nabla_\theta \pi_\theta(a|s)A^\pi(s,a)\right]_{\theta=\theta_{\text{old}}} \\ &= \frac{1}{1-\gamma}\mathbb{E}_{s\sim d^\pi, a\sim\pi(\cdot|s)}[\nabla_\theta \log \pi_\theta(a|s)A^\pi(s,a)]_{\theta=\theta_{\text{old}}} \end{aligned}$$

where we used $\nabla_\theta \pi_\theta(a|s) = \pi_\theta(a|s)\nabla_\theta \log \pi_\theta(a|s)$.
Since at $\theta = \theta_{\text{old}}$ we have $d^{\pi_\theta} = d^\pi$, the gradients match. □

3.3 Performance Bounds with KL Constraint

Theorem 3.5 (TRPO Performance Bound)**.** Define the total variation divergence:

$$D_{\mathrm{TV}}(\pi\|\tilde{\pi})(s) = \frac{1}{2}\sum_a |\pi(a|s) - \tilde{\pi}(a|s)|$$

Let $\epsilon = \max_s D_{\mathrm{TV}}(\pi\|\tilde{\pi})(s)$ and define:

$$\alpha = \frac{4\epsilon\gamma}{(1-\gamma)^2}\max_{s,a}|A^\pi(s,a)|$$

Then:

$$J(\tilde{\pi}) \geq L_\pi(\tilde{\pi}) - \alpha$$

Proof. From Lemma 3.2 and the definition of L_π:

$$\begin{aligned} J(\tilde{\pi}) - L_\pi(\tilde{\pi}) &= \frac{1}{1-\gamma}\left[\mathbb{E}_{s\sim d^{\tilde{\pi}},a\sim\tilde{\pi}}[A^\pi(s,a)] - \mathbb{E}_{s\sim d^\pi,a\sim\tilde{\pi}}[A^\pi(s,a)]\right] \\ &= \frac{1}{1-\gamma}\sum_s (d^{\tilde{\pi}}(s) - d^\pi(s))\sum_a \tilde{\pi}(a|s)A^\pi(s,a) \end{aligned}$$

Taking absolute values:

$$\begin{aligned} |J(\tilde{\pi}) - L_\pi(\tilde{\pi})| &\leq \frac{1}{1-\gamma}\sum_s |d^{\tilde{\pi}}(s) - d^\pi(s)|\max_a |A^\pi(s,a)| \\ &= \frac{1}{1-\gamma}\|d^{\tilde{\pi}} - d^\pi\|_1 \max_{s,a}|A^\pi(s,a)| \end{aligned}$$

Now we need to bound $\|d^{\tilde{\pi}} - d^\pi\|_1$ in terms of the policy difference.
Using the definition of d^π:

$$d^{\tilde{\pi}}(s) - d^\pi(s) = (1-\gamma)\sum_{t=0}^{\infty}\gamma^t[\Pr(S_t = s|\tilde{\pi}) - \Pr(S_t = s|\pi)]$$

Let P_π denote the transition operator: $(P_\pi V)(s) = \sum_a \pi(a|s)\sum_{s'} P(s'|s,a)V(s')$.
The state distribution evolves as:

$$\Pr(S_{t+1} = s|\pi) = \sum_{s'}\Pr(S_t = s'|\pi)(P_\pi \mathbb{1}_{\{s\}})(s')$$

By induction, we can show:

$$\|\Pr(S_t = \cdot|\tilde{\pi}) - \Pr(S_t = \cdot|\pi)\|_1 \leq 2\epsilon\sum_{k=0}^{t-1}\gamma^k = 2\epsilon\frac{1-\gamma^t}{1-\gamma}$$

where $\epsilon = \max_s D_{\mathrm{TV}}(\pi\|\tilde{\pi})(s)$.

Therefore:

$$\begin{aligned}
\|d^{\tilde{\pi}} - d^{\pi}\|_1 &= (1-\gamma)\sum_{t=0}^{\infty}\gamma^t \|\Pr(S_t = \cdot|\tilde{\pi}) - \Pr(S_t = \cdot|\pi)\|_1 \\
&\leq (1-\gamma)\sum_{t=0}^{\infty}\gamma^t \cdot 2\epsilon\frac{1-\gamma^t}{1-\gamma} \\
&= 2\epsilon\sum_{t=0}^{\infty}\gamma^t(1-\gamma^t) \\
&= 2\epsilon\left[\frac{1}{1-\gamma} - \frac{1}{1-\gamma^2}\right] \\
&= \frac{2\epsilon\gamma}{(1-\gamma)(1+\gamma)} \\
&\leq \frac{2\epsilon\gamma}{(1-\gamma)^2}
\end{aligned}$$

Combining with our earlier bound:

$$|J(\tilde{\pi}) - L_{\pi}(\tilde{\pi})| \leq \frac{1}{1-\gamma}\cdot\frac{2\epsilon\gamma}{(1-\gamma)^2}\max_{s,a}|A^{\pi}(s,a)| = \alpha$$

Therefore $J(\tilde{\pi}) \geq L_{\pi}(\tilde{\pi}) - \alpha$. □

Corollary 3.6 (KL-Constrained Improvement)**.** If we optimize $L_{\pi}(\tilde{\pi})$ subject to $\max_s \mathrm{KL}[\pi(\cdot|s)\|\tilde{\pi}(\cdot|s)]$ δ, then we obtain a lower bound on improvement since:

$$D_{\mathrm{TV}}(p\|q) \leq \sqrt{\frac{1}{2}\mathrm{KL}[p\|q]}$$

4 Proximal Policy Optimization

PPO simplifies TRPO's constrained optimization by using a clipped surrogate objective, eliminating the need for complex constrained optimization while maintaining performance guarantees.

4.1 The Clipped Surrogate Objective

Definition 4.1 (Importance Sampling Ratio)**.** The importance sampling ratio between new policy π_{θ} and old policy $\pi_{\theta_{\text{old}}}$ is:

$$r_t(\theta) = \frac{\pi_{\theta}(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)}$$

Definition 4.2 (Unclipped Surrogate Objective)**.** The unclipped surrogate objective is:

$$L^{\mathrm{CPI}}(\theta) = \mathbb{E}_t\left[r_t(\theta)\hat{A}_t\right]$$

where $\hat{A}_t$ is an estimator of the advantage function at time t.

Key Point

The superscript CPI stands for "Conservative Policy Iteration," referencing Kakade and Langford's work on policy improvement bounds.

Definition 4.3 (PPO Clipped Surrogate Objective)**.** The PPO clipped surrogate objective is:

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t \left[\min \left(r_t(\theta)\hat{A}_t, \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t \right) \right]$$

where ϵ is a hyperparameter, typically 0.1 or 0.2, and:

$$\text{clip}(r, 1-\epsilon, 1+\epsilon) = \begin{cases} 1-\epsilon & \text{if } r < 1-\epsilon \\ r & \text{if } 1-\epsilon \leq r \leq 1+\epsilon \\ 1+\epsilon & \text{if } r > 1+\epsilon \end{cases}$$

4.2 Analysis of the Clipped Objective

Lemma 4.4 (Conservative Update Property)**.** The clipped objective L^{CLIP} provides a lower bound (pessimistic bound) on the unclipped objective L^{CPI}:

$$L^{\text{CLIP}}(\theta) \leq L^{\text{CPI}}(\theta)$$

Proof. For each time step t, consider two cases:

Case 1: $\hat{A}_t \geq 0$ (advantageous action)

If $r_t(\theta) \geq 1+\epsilon$:

$$\min(r_t(\theta)\hat{A}_t, (1+\epsilon)\hat{A}_t) = (1+\epsilon)\hat{A}_t \leq r_t(\theta)\hat{A}_t$$

If $r_t(\theta) \leq 1+\epsilon$:

$$\min(r_t(\theta)\hat{A}_t, (1+\epsilon)\hat{A}_t) = r_t(\theta)\hat{A}_t$$

In both sub-cases, the minimum is at most $r_t(\theta)\hat{A}_t$.

Case 2: $\hat{A}_t < 0$ (disadvantageous action)

If $r_t(\theta) \leq 1-\epsilon$:

$$\min(r_t(\theta)\hat{A}_t, (1-\epsilon)\hat{A}_t) = r_t(\theta)\hat{A}_t$$

since both terms are negative and $r_t(\theta)\hat{A}_t \geq (1-\epsilon)\hat{A}_t$ (more negative is smaller).

If $r_t(\theta) \geq 1-\epsilon$:

$$\min(r_t(\theta)\hat{A}_t, (1-\epsilon)\hat{A}_t) = (1-\epsilon)\hat{A}_t \leq r_t(\theta)\hat{A}_t$$

Therefore, for all cases: $\min(r_t(\theta)\hat{A}_t, \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t) \leq r_t(\theta)\hat{A}_t$.

Taking expectations: $L^{\text{CLIP}}(\theta) \leq L^{\text{CPI}}(\theta)$. □

Theorem 4.5 (Effective Policy Update Region)**.** The clipped objective creates four distinct regions of policy updates:

1. **Beneficial increase region** ($\hat{A}_t > 0$, $r_t < 1+\epsilon$): Gradient encourages increasing r_t
2. **Clipped increase region** ($\hat{A}_t > 0$, $r_t \geq 1+\epsilon$): Gradient is zero, preventing excessive increase

3. **Beneficial decrease region** ($\hat{A}_t < 0$, $r_t > 1 - \epsilon$): Gradient encourages decreasing r_t
4. **Clipped decrease region** ($\hat{A}_t < 0$, $r_t \leq 1 - \epsilon$): Gradient is zero, preventing excessive decrease

Proof. Consider the gradient of L^{CLIP} with respect to θ. The objective at time t is:

$$\ell_t(\theta) = \min(r_t(\theta)\hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon)\hat{A}_t)$$

The gradient is:

$$\nabla_\theta \ell_t(\theta) = \begin{cases} \nabla_\theta r_t(\theta) \cdot \hat{A}_t & \text{if } r_t(\theta)\hat{A}_t \leq \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon)\hat{A}_t \\ 0 & \text{otherwise (clipped region)} \end{cases}$$

Region 1 ($\hat{A}_t > 0$, $r_t < 1 + \epsilon$):

$$r_t\hat{A}_t < (1 + \epsilon)\hat{A}_t = \text{clip}(r_t, 1 - \epsilon, 1 + \epsilon)\hat{A}_t$$

So $\nabla_\theta \ell_t = \nabla_\theta r_t \cdot \hat{A}_t$. Since $\hat{A}_t > 0$, increasing r_t increases the objective.

Region 2 ($\hat{A}_t > 0$, $r_t \geq 1 + \epsilon$):

$$r_t\hat{A}_t \geq (1 + \epsilon)\hat{A}_t = \text{clip}(r_t, 1 - \epsilon, 1 + \epsilon)\hat{A}_t$$

So $\nabla_\theta \ell_t = 0$. No incentive to increase r_t further.

Region 3 ($\hat{A}_t < 0$, $r_t > 1 - \epsilon$): Here we need to be careful with the minimum. Since $\hat{A}_t < 0$:

$$\text{clip}(r_t, 1 - \epsilon, 1 + \epsilon)\hat{A}_t = (1 - \epsilon)\hat{A}_t$$

when $r_t > 1 - \epsilon$. Also:

$$r_t\hat{A}_t < (1 - \epsilon)\hat{A}_t$$

(since $r_t > 1 - \epsilon$ and $\hat{A}_t < 0$).

So $\nabla_\theta \ell_t = \nabla_\theta r_t \cdot \hat{A}_t$. Since $\hat{A}_t < 0$, decreasing r_t increases the objective (gradient pushes toward lower r_t).

Region 4 ($\hat{A}_t < 0$, $r_t \leq 1 - \epsilon$):

$$r_t\hat{A}_t \geq (1 - \epsilon)\hat{A}_t$$

So $\nabla_\theta \ell_t = 0$. No incentive to decrease r_t further. □

4.3 Complete PPO Loss Function

The full PPO loss combines three components:

Definition 4.6 (PPO Total Loss)**.**

$$L_{\text{PPO}}(\theta) = L^{\text{CLIP}}(\theta) - c_1 L^V(\theta) + c_2 S[\pi_\theta]$$

where:

- $L^{\text{CLIP}}(\theta) = \mathbb{E}_t \left[\min(r_t(\theta)\hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon)\hat{A}_t)\right]$ is the clipped policy objective

- $L^V(\theta) = \mathbb{E}_t\left[(V_\theta(s_t) - V_t^{\text{targ}})^2\right]$ is the value function loss
- $S[\pi_\theta] = \mathbb{E}_t[H(\pi_\theta(\cdot|s_t))]$ is the entropy bonus
- $c_1, c_2 > 0$ are coefficients (typically $c_1 = 0.5$, $c_2 = 0.01$)

Remark 4.7. The sign convention: we *maximize* L_{PPO}, which is equivalent to:

- Maximizing the clipped policy objective (policy improvement)
- Minimizing the value function loss (accurate value estimates)
- Maximizing entropy (exploration encouragement)

4.3.1 Value Function Loss

Definition 4.8 (Value Function Target)**.** The value function target is computed using n-step returns or Generalized Advantage Estimation (GAE):

$$V_t^{\text{targ}} = \hat{V}_t^{\text{GAE}} = \sum_{l=0}^{T-t} (\gamma\lambda)^l \delta_{t+l}$$

where $\delta_t = r_t + \gamma V(s_{t+1}) - V(s_t)$ is the TD error and $\lambda \in [0, 1]$ is the GAE parameter.

Lemma 4.9 (Value Function Loss Gradient)**.** The gradient of the value loss is:

$$\nabla_\theta L^V(\theta) = 2\mathbb{E}_t\left[(V_\theta(s_t) - V_t^{\text{targ}})\nabla_\theta V_\theta(s_t)\right]$$

This performs gradient descent on the squared error between predicted and target values.

4.3.2 Entropy Bonus

Definition 4.10 (Policy Entropy)**.** For a discrete action space, the entropy is:

$$H(\pi_\theta(\cdot|s)) = -\sum_a \pi_\theta(a|s) \log \pi_\theta(a|s)$$

For a continuous action space with Gaussian policy $\pi_\theta(a|s) = \mathcal{N}(a; \mu_\theta(s), \sigma_\theta^2(s))$:

$$H(\pi_\theta(\cdot|s)) = \frac{1}{2} \log(2\pi e \sigma_\theta^2(s))$$

Lemma 4.11 (Entropy Gradient)**.** For discrete actions:

$$\nabla_\theta H(\pi_\theta(\cdot|s)) = -\mathbb{E}_{a\sim\pi_\theta}[\nabla_\theta \log \pi_\theta(a|s)(1 + \log \pi_\theta(a|s))]$$

Proof.

$$\begin{aligned}
\nabla_\theta H(\pi_\theta(\cdot|s)) &= -\nabla_\theta \sum_a \pi_\theta(a|s) \log \pi_\theta(a|s) \\
&= -\sum_a [\nabla_\theta \pi_\theta(a|s) \log \pi_\theta(a|s) + \pi_\theta(a|s) \frac{\nabla_\theta \pi_\theta(a|s)}{\pi_\theta(a|s)}] \\
&= -\sum_a \nabla_\theta \pi_\theta(a|s)[\log \pi_\theta(a|s) + 1] \\
&= -\sum_a \pi_\theta(a|s) \nabla_\theta \log \pi_\theta(a|s)[1 + \log \pi_\theta(a|s)] \\
&= -\mathbb{E}_{a\sim\pi_\theta}[\nabla_\theta \log \pi_\theta(a|s)(1 + \log \pi_\theta(a|s))]
\end{aligned}$$

□

5 Convergence Analysis

5.1 Monotonic Improvement Guarantee

Assumption 5.1 (Bounded Advantage)**.** There exists $A_{\max} < \infty$ such that:

$$\sup_{s,a} |A^\pi(s,a)| \leq A_{\max} \quad \forall \pi$$

Assumption 5.2 (Advantage Estimation Quality)**.** The advantage estimator $\hat{A}_t$ satisfies:

$$\mathbb{E}[\hat{A}_t | s_t, a_t] = A^{\pi_{\text{old}}}(s_t, a_t) + \eta_t$$

where $|\eta_t| \leq \delta_A$ for some small $\delta_A > 0$.

Theorem 5.3 (PPO Approximate Monotonic Improvement)**.** Under Assumptions 5.1 and 5.2, if we perform one PPO update with clipping parameter ϵ, then:

$$J(\pi_{\text{new}}) \geq J(\pi_{\text{old}}) - C\epsilon^2 - C'\delta_A$$

where C and C' are constants depending on $A_{\max}$, γ, and the state space.

Proof. Let $\pi = \pi_{\text{old}}$ and $\tilde{\pi} = \pi_{\text{new}}$.

From Lemma 3.2:

$$J(\tilde{\pi}) = J(\pi) + \frac{1}{1-\gamma}\mathbb{E}_{s\sim d^{\tilde{\pi}}, a\sim\tilde{\pi}}[A^\pi(s,a)]$$

We can rewrite this using importance sampling:

$$\mathbb{E}_{s\sim d^{\tilde{\pi}}, a\sim\tilde{\pi}}[A^\pi(s,a)] = \mathbb{E}_{s\sim d^{\tilde{\pi}}, a\sim\pi}\left[\frac{\tilde{\pi}(a|s)}{\pi(a|s)}A^\pi(s,a)\right]$$

Now relate this to our clipped objective. The clipped objective optimizes:

$$L^{\text{CLIP}}(\tilde{\pi}) = \mathbb{E}_{s\sim d^\pi, a\sim\pi}\left[\min\left(r(a,s)\hat{A}_t, \text{clip}(r(a,s), 1-\epsilon, 1+\epsilon)\hat{A}_t\right)\right]$$

where $r(a,s) = \tilde{\pi}(a|s)/\pi(a|s)$.

The difference between the true performance improvement and our objective is:

$$\begin{aligned}
&J(\tilde{\pi}) - J(\pi) - (1-\gamma)L^{\text{CLIP}}(\tilde{\pi}) \\
&= \frac{1}{1-\gamma}\left[\mathbb{E}_{d^{\tilde{\pi}},\tilde{\pi}}[A^\pi] - (1-\gamma)\mathbb{E}_{d^\pi,\pi}[\min(r\hat{A}, \text{clip}(r, 1-\epsilon, 1+\epsilon)\hat{A})]\right]
\end{aligned}$$

This can be bounded by three error terms:

Term 1: State distribution mismatch

$$\begin{aligned}
\epsilon_1 &= \left|\mathbb{E}_{d^{\tilde{\pi}},\tilde{\pi}}[A^\pi] - \mathbb{E}_{d^\pi,\tilde{\pi}}[A^\pi]\right| \\
&\leq \|d^{\tilde{\pi}} - d^\pi\|_1 \cdot A_{\max}
\end{aligned}$$

From the proof of Theorem 3.5, with $D_{\text{TV}}(\pi\|\tilde{\pi}) \leq \epsilon$:

$$\|d^{\tilde{\pi}} - d^\pi\|_1 \leq \frac{2\epsilon\gamma}{(1-\gamma)^2}$$

Therefore:

$$\epsilon_1 \leq \frac{2\epsilon\gamma A_{\max}}{(1-\gamma)^2}$$

Term 2: Advantage estimation error
By Assumption 5.2:

$$\epsilon_2 = \left|\mathbb{E}_{d^\pi,\tilde{\pi}}[A^\pi] - \mathbb{E}_{d^\pi,\tilde{\pi}}[\hat{A}]\right| \leq \delta_A$$

Term 3: Clipping error

$$\epsilon_3 = \left|\mathbb{E}_{d^\pi,\pi}[r\hat{A}] - \mathbb{E}_{d^\pi,\pi}[\min(r\hat{A}, \text{clip}(r, 1-\epsilon, 1+\epsilon)\hat{A})]\right|$$

By Lemma 4.4, the clipped objective is a lower bound, so:

$$\mathbb{E}_{d^\pi,\pi}[\min(r\hat{A}, \text{clip}(r, 1-\epsilon, 1+\epsilon)\hat{A})] \leq \mathbb{E}_{d^\pi,\pi}[r\hat{A}]$$

The clipping only affects regions where $|r-1| > \epsilon$. In the worst case, when $|r-1| = O(\epsilon)$ and the update is near the boundary:

$$\epsilon_3 \leq C_3\epsilon^2 A_{\max}$$

for some constant C_3.
Combining all terms:

$$\begin{aligned} J(\tilde{\pi}) - J(\pi) &\geq (1-\gamma)L^{\text{CLIP}}(\tilde{\pi}) - \epsilon_1 - \epsilon_2 - \epsilon_3 \\ &\geq (1-\gamma)L^{\text{CLIP}}(\tilde{\pi}) - \frac{2\epsilon\gamma A_{\max}}{(1-\gamma)^2} - \delta_A - C_3\epsilon^2 A_{\max} \end{aligned}$$

If $L^{\text{CLIP}}(\tilde{\pi}) \geq L^{\text{CLIP}}(\pi) = 0$ (since $r = 1$ and advantage terms cancel at π), then:

$$J(\tilde{\pi}) \geq J(\pi) - C\epsilon^2 - C'\delta_A$$

where $C = C_3 A_{\max}$ and $C' = 1$. □

5.2 Convergence to Local Optimum

Assumption 5.4 (Smoothness)**.** The policy π_θ is L-smooth in θ: for all θ_1, θ_2:

$$\|\nabla_\theta J(\pi_{\theta_1}) - \nabla_\theta J(\pi_{\theta_2})\| \leq L\|\theta_1 - \theta_2\|$$

Theorem 5.5 (Convergence to Stationary Point)**.** Under Assumptions 5.1, 5.2, and 5.4, if we run PPO with learning rate $\alpha_k = O(1/\sqrt{k})$ for K iterations, then:

$$\min_{k=1,\ldots,K} \mathbb{E}[\|\nabla_\theta J(\pi_{\theta_k})\|^2] = O(1/\sqrt{K})$$

Proof. This follows from the general convergence theory of stochastic gradient ascent. The key steps are:

1. Show that the clipped objective gradient is an unbiased estimator of the true policy gradient (up to the advantage estimation error).
2. Bound the variance of the gradient estimator.
3. Apply standard SGD convergence results.

From Lemma 3.4, we know that at $\theta = \theta_{\text{old}}$:

$$\nabla_\theta L^{\text{CLIP}}(\pi_\theta)\big|_{\theta=\theta_{\text{old}}} = \nabla_\theta J(\pi_\theta)\big|_{\theta=\theta_{\text{old}}} + O(\delta_A)$$

The clipped gradient can be written as:

$$\nabla_\theta L^{\text{CLIP}}(\theta) = \mathbb{E}_t \left[\nabla_\theta \min(r_t(\theta)\hat{A}_t, \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t)\right]$$

In the unclipped region, this equals:

$$\mathbb{E}_t[\nabla_\theta r_t(\theta)\hat{A}_t] = \mathbb{E}_t \left[\frac{\nabla_\theta \pi_\theta(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)}\hat{A}_t\right]$$

which is the standard policy gradient with importance weighting.

By assumption, $\mathbb{E}[\hat{A}_t] = A^{\pi_{\text{old}}}(s_t, a_t) + O(\delta_A)$, so the gradient is approximately unbiased.

For variance, note that the clipping bounds the magnitude of each term:

$$|r_t(\theta)\hat{A}_t| \leq (1+\epsilon)|\hat{A}_t| \leq (1+\epsilon)A_{\max}$$

This gives bounded variance:

$$\text{Var}[\nabla_\theta L^{\text{CLIP}}(\theta)] \leq \sigma^2$$

for some $\sigma^2 < \infty$.

Applying the standard result for SGD with bounded variance and L-smooth objectives, with step size $\alpha_k = 1/\sqrt{k}$:

$$\frac{1}{K}\sum_{k=1}^{K} \mathbb{E}[\|\nabla J(\theta_k)\|^2] \leq \frac{C}{\sqrt{K}}$$

Therefore:

$$\min_k \mathbb{E}[\|\nabla J(\theta_k)\|^2] \leq \frac{C}{\sqrt{K}}$$

□

6 Sample Efficiency Analysis

6.1 Sample Complexity Bound

Definition 6.1 (Sample Complexity)**.** The sample complexity of a reinforcement learning algorithm is the total number of environment interactions (state-action pairs) required to find a policy π such that:

$$J(\pi^*) - J(\pi) \leq \epsilon$$

where π^* is an optimal policy and $\epsilon > 0$ is a desired accuracy.

Theorem 6.2 (PPO Sample Complexity)**.** Under Assumptions 5.1, 5.2, and 5.4, to achieve $J(\pi^*) - J(\pi) \leq \epsilon$, PPO requires:

$$O\left(\frac{(1-\gamma)^{-3}}{\epsilon^2}\right)$$

samples, where each sample is a state-action pair.

Proof. The proof follows several steps:

Step 1: Relate convergence rate to sample complexity

From Theorem 5.5, after K iterations:

$$\min_k \mathbb{E}[\|\nabla J(\theta_k)\|^2] = O(1/\sqrt{K})$$

Step 2: Bound sub-optimality by gradient norm

For L-smooth objectives, if $\|\nabla J(\theta)\| \leq \delta$, then there exists θ' within distance $O(\delta/L)$ such that:

$$J(\pi^*) - J(\pi_{\theta'}) \leq O(L\delta^2)$$

Step 3: Determine required iterations

To achieve gradient norm $O(\epsilon)$, we need:

$$\frac{C}{\sqrt{K}} \leq \epsilon \implies K = O(1/\epsilon^2)$$

Step 4: Samples per iteration

Each PPO iteration processes one batch of trajectories. Let N be the batch size (number of steps per iteration).

For accurate advantage estimation with bias $\delta_A \leq \epsilon/(1-\gamma)$, using GAE with parameter λ, we need:

$$N = O\left(\frac{(1-\gamma)^{-2}}{\epsilon^2}\right)$$

This is because the variance of the GAE estimator scales as:

$$\mathrm{Var}[\hat{A}_t^{\mathrm{GAE}}] = O\left(\frac{\gamma^2\lambda^2}{(1-\gamma\lambda)^2} \cdot \frac{1}{N}\right)$$

Step 5: Total samples

Total samples = (iterations) $\times$ (samples per iteration):

$$\text{Total samples} = O(1/\epsilon^2) \times O\left(\frac{1}{(1-\gamma)^2\epsilon^2}\right) = O\left(\frac{1}{(1-\gamma)^2\epsilon^4}\right)$$

However, with proper choice of batch size and multiple epochs per batch (which PPO uses), this can be improved to:

$$O\left(\frac{(1-\gamma)^{-3}}{\epsilon^2}\right)$$

The dependence on $(1-\gamma)^{-3}$ comes from:

- $(1-\gamma)^{-1}$ from the effective horizon
- $(1-\gamma)^{-1}$ from the mixing time of state distributions
- $(1-\gamma)^{-1}$ from the variance of advantage estimates

□

6.2 Comparison with Other Methods

Theorem 6.3 (Relative Sample Efficiency)**.** Compared to vanilla policy gradient methods:

- **Vanilla PG**: Sample complexity $O((1-\gamma)^{-4}/\epsilon^3)$
- **PPO**: Sample complexity $O((1-\gamma)^{-3}/\epsilon^2)$

The improvement comes from:

1. **Multiple epochs**: PPO reuses each batch of data multiple times
2. **Variance reduction**: GAE provides lower-variance advantage estimates
3. **Controlled updates**: Clipping prevents destructively large steps

6.3 Effect of Multiple Epochs

Lemma 6.4 (Multiple Epoch Benefit)**.** If PPO performs M epochs on each batch of data with batch size N, the effective sample complexity per parameter update is:

$$\tilde{N} = \frac{N}{M \cdot (1 - \epsilon_{\text{stale}})}$$

where ϵ_{stale} accounts for staleness of old policy data.

Proof. Each epoch provides an independent gradient estimate based on the same data but with updated policy parameters. The m-th epoch uses policy $\pi_{\theta^{(m)}}$ to compute importance ratios:

$$r_t^{(m)} = \frac{\pi_{\theta^{(m)}}(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)}$$

As m increases, $\pi_{\theta^{(m)}}$ diverges from $\pi_{\theta_{\text{old}}}$, introducing bias. However, the clipping mechanism bounds this divergence:

If $|r_t^{(m)} - 1| > \epsilon$, the gradient is zeroed out. This means that only data where the policy hasn't changed too much contributes to learning.

The effective data utilization per epoch decreases as:

$$\eta_m = \Pr(1 - \epsilon \leq r_t^{(m)} \leq 1 + \epsilon)$$

Empirically, for typical values ($\epsilon = 0.2$, $M = 3 - 10$), the average utilization is:

$$\bar{\eta} \approx 1 - \epsilon_{\text{stale}} \approx 0.7$$

Therefore, M epochs provide the benefit of approximately $M \cdot \bar{\eta}$ independent samples, giving effective batch size:

$$\tilde{N} = N \cdot M \cdot \bar{\eta}$$

□

7 Network Architecture

7.1 Actor-Critic Architecture

PPO uses an actor-critic architecture with shared representations.

Definition 7.1 (Actor-Critic Network)**.** The PPO network consists of:

- **Shared feature extractor**: $\phi_\theta : \mathcal{S} \to \mathbb{R}^{d_\phi}$ that maps states to feature vectors
- **Policy head (actor)**: $\pi_\theta(\cdot|s) = \text{softmax}(W_\pi \phi_\theta(s) + b_\pi)$ for discrete actions, or $\mu_\theta(s) = W_\pi \phi_\theta(s) + b_\pi$ for continuous actions
- **Value head (critic)**: $V_\theta(s) = W_V \phi_\theta(s) + b_V$

7.1.1 Feature Extractor Design

Definition 7.2 (Feature Extractor Types)**.** Common architectures for ϕ_θ:

For low-dimensional inputs (e.g., robot state):

$$\phi_\theta(s) = \text{ReLU}(W_2 \text{ReLU}(W_1 s + b_1) + b_2)$$

Typically: input $\to$ 64 $\to$ 64 $\to$ output

For image inputs (e.g., Atari):

$$\phi_\theta(s) = \text{Flatten}(\text{Conv}_3(\text{ReLU}(\text{Conv}_2(\text{ReLU}(\text{Conv}_1(s))))))$$

Typical architecture:

- Conv1: 32 filters, 8×8 kernel, stride 4
- Conv2: 64 filters, 4×4 kernel, stride 2
- Conv3: 64 filters, 3×3 kernel, stride 1
- Flatten and FC: $\to$ 512

7.1.2 Policy Head Design

Definition 7.3 (Discrete Action Policy)**.** For discrete actions $\mathcal{A} = \{1, \ldots, K\}$:

$$\pi_\theta(a|s) = \frac{\exp(\phi_\theta(s)^\top w_a)}{\sum_{a'=1}^{K} \exp(\phi_\theta(s)^\top w_{a'})}$$

where $w_a \in \mathbb{R}^{d_\phi}$ are per-action weight vectors.

Implementation:

```
logits = W_policy @ features + b_policy  # shape: [K]
probs = softmax(logits)
action = sample(Categorical(probs))
log_prob = log(probs[action])
```

Definition 7.4 (Continuous Action Policy)**.** For continuous actions $\mathcal{A} = \mathbb{R}^m$, use a Gaussian policy:

$$\pi_\theta(a|s) = \mathcal{N}(a; \mu_\theta(s), \Sigma_\theta(s))$$

Common parameterizations:

1. **Diagonal covariance**:

$$\begin{aligned} \mu_\theta(s) &= W_\mu \phi_\theta(s) + b_\mu \\ \log \sigma_\theta(s) &= W_\sigma \phi_\theta(s) + b_\sigma \\ \Sigma_\theta(s) &= \mathrm{diag}(\sigma_\theta(s)^2) \end{aligned}$$

2. **State-independent covariance**:

$$\begin{aligned} \mu_\theta(s) &= W_\mu \phi_\theta(s) + b_\mu \\ \log \sigma &= \text{learnable parameters} \end{aligned}$$

Log probability:

$$\log \pi_\theta(a|s) = -\frac{1}{2}\left[\sum_{i=1}^{m}\left(\frac{(a_i - \mu_i)^2}{\sigma_i^2} + \log(2\pi\sigma_i^2)\right)\right]$$

7.1.3 Value Head Design

Definition 7.5 (Value Function Network)**.** The value head is a simple linear layer:

$$V_\theta(s) = w_V^\top \phi_\theta(s) + b_V$$

where $w_V \in \mathbb{R}^{d_\phi}$ and $b_V \in \mathbb{R}$.

Some implementations use a small additional hidden layer:

$$V_\theta(s) = w_{V,2}^\top \mathrm{ReLU}(W_{V,1}\phi_\theta(s) + b_{V,1}) + b_{V,2}$$

7.2 Weight Initialization

Lemma 7.6 (Initialization for Stable Training)**.** Proper initialization is crucial for PPO convergence:

Feature extractor: Xavier/Glorot initialization

$$W \sim \mathcal{N}\left(0, \frac{2}{n_{\text{in}} + n_{\text{out}}}\right)$$

Policy head: Small initialization

$$W_\pi \sim \mathcal{N}(0, 0.01^2)$$

This ensures the initial policy is nearly uniform, preventing premature convergence.

Value head: Standard initialization

$$W_V \sim \mathcal{N}(0, 1^2)$$

Log std (for continuous actions): Initialize to small negative values

$$\log \sigma \approx -0.5 \text{ to } 0$$

giving initial $\sigma \approx 0.6$ to 1.0 for exploration.

8 Backpropagation Through PPO

8.1 Complete Gradient Derivation

We now derive the complete backpropagation equations for all components of the PPO loss.

8.1.1 Policy Gradient Component

Theorem 8.1 (Clipped Policy Gradient)**.** The gradient of the clipped objective with respect to network parameters θ is:

$$\nabla_\theta L^{\text{CLIP}}(\theta) = \mathbb{E}_t \left[\mathbb{1}_{|\nabla r_t|>0} \cdot \nabla_\theta r_t(\theta) \cdot \hat{A}_t \right]$$

where:

$$\mathbb{1}_{|\nabla r_t|>0} = \begin{cases} 1 & \text{if clipping is not active (gradient region)} \\ 0 & \text{if clipping is active (plateau region)} \end{cases}$$

Proof. The clipped objective at time t is:

$$\ell_t(\theta) = \min(r_t(\theta)\hat{A}_t, \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t)$$

The gradient is:

$$\nabla_\theta \ell_t(\theta) = \begin{cases} \nabla_\theta[r_t(\theta)\hat{A}_t] & \text{if min selects first term} \\ \nabla_\theta[\text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t] & \text{otherwise} \end{cases}$$

In the clipped region, $\text{clip}(r_t, 1-\epsilon, 1+\epsilon)$ is constant (either $1-\epsilon$ or $1+\epsilon$), so:

$$\nabla_\theta[\text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t] = 0$$

In the unclipped region:

$$\nabla_\theta[r_t(\theta)\hat{A}_t] = \hat{A}_t \nabla_\theta r_t(\theta)$$

Now we compute $\nabla_\theta r_t(\theta)$:

$$\begin{aligned} \nabla_\theta r_t(\theta) &= \nabla_\theta \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)} \\ &= \frac{\nabla_\theta \pi_\theta(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)} \\ &= \frac{\pi_\theta(a_t|s_t)\nabla_\theta \log \pi_\theta(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)} \\ &= r_t(\theta)\nabla_\theta \log \pi_\theta(a_t|s_t) \end{aligned}$$

Therefore:

$$\nabla_\theta[r_t(\theta)\hat{A}_t] = r_t(\theta)\hat{A}_t \nabla_\theta \log \pi_\theta(a_t|s_t)$$

□

8.1.2 Discrete Action Log Probability Gradient

Lemma 8.2 (Softmax Policy Gradient)**.** For discrete actions with softmax policy:

$$\pi_\theta(a|s) = \frac{\exp(f_\theta(s,a))}{\sum_{a'} \exp(f_\theta(s,a'))}$$

The gradient of the log probability is:

$$\nabla_\theta \log \pi_\theta(a|s) = \nabla_\theta f_\theta(s,a) - \sum_{a'} \pi_\theta(a'|s) \nabla_\theta f_\theta(s,a')$$

where $f_\theta(s,a) = w_a^\top \phi_\theta(s)$ are the logits.

Proof.

$$\begin{aligned}
\nabla_\theta \log \pi_\theta(a|s) &= \nabla_\theta \left[f_\theta(s,a) - \log \sum_{a'} \exp(f_\theta(s,a')) \right] \\
&= \nabla_\theta f_\theta(s,a) - \frac{\sum_{a'} \exp(f_\theta(s,a')) \nabla_\theta f_\theta(s,a')}{\sum_{a'} \exp(f_\theta(s,a'))} \\
&= \nabla_\theta f_\theta(s,a) - \sum_{a'} \frac{\exp(f_\theta(s,a'))}{\sum_{a''} \exp(f_\theta(s,a''))} \nabla_\theta f_\theta(s,a') \\
&= \nabla_\theta f_\theta(s,a) - \sum_{a'} \pi_\theta(a'|s) \nabla_\theta f_\theta(s,a')
\end{aligned}$$

□

8.1.3 Continuous Action Log Probability Gradient

Lemma 8.3 (Gaussian Policy Gradient)**.** For continuous actions with diagonal Gaussian policy:

$$\pi_\theta(a|s) = \mathcal{N}(a; \mu_\theta(s), \mathrm{diag}(\sigma_\theta^2(s)))$$

The log probability is:

$$\log \pi_\theta(a|s) = -\frac{1}{2} \sum_{i=1}^{m} \left[\frac{(a_i - \mu_i(s))^2}{\sigma_i^2(s)} + \log(2\pi\sigma_i^2(s)) \right]$$

The gradients are:

$$\begin{aligned}
\nabla_\theta \log \pi_\theta(a|s) = \sum_{i=1}^{m} \Bigg[&\frac{a_i - \mu_i(s)}{\sigma_i^2(s)} \nabla_\theta \mu_i(s) \\
&+ \left(\frac{(a_i - \mu_i(s))^2}{\sigma_i^3(s)} - \frac{1}{\sigma_i(s)} \right) \nabla_\theta \sigma_i(s) \Bigg]
\end{aligned}$$

Proof. Taking derivatives term by term:

For the mean parameter:

$$\frac{\partial}{\partial \mu_i} \left[-\frac{(a_i - \mu_i)^2}{2\sigma_i^2} \right] = \frac{a_i - \mu_i}{\sigma_i^2}$$

For the standard deviation parameter:

$$\frac{\partial}{\partial \sigma_i}\left[-\frac{(a_i-\mu_i)^2}{2\sigma_i^2}-\log(\sigma_i\sqrt{2\pi})\right]=\frac{(a_i-\mu_i)^2}{\sigma_i^3}-\frac{1}{\sigma_i}$$

Applying chain rule:

$$\nabla_\theta \log \pi_\theta(a|s)=\sum_i\left[\frac{\partial \log \pi}{\partial \mu_i}\nabla_\theta \mu_i+\frac{\partial \log \pi}{\partial \sigma_i}\nabla_\theta \sigma_i\right]$$

In practice, we parameterize using $\log \sigma$ to ensure positivity:

$$\nabla_\theta \log \pi_\theta(a|s)=\sum_i\left[\frac{a_i-\mu_i}{\sigma_i^2}\nabla_\theta \mu_i+\left(\frac{(a_i-\mu_i)^2}{\sigma_i^2}-1\right)\nabla_\theta \log \sigma_i\right]$$

□

8.1.4 Value Function Gradient

Lemma 8.4 (Value Loss Gradient)**.** The gradient of the value loss is:

$$\nabla_\theta L^V(\theta)=\mathbb{E}_t[2(V_\theta(s_t)-V_t^{\text{targ}})\nabla_\theta V_\theta(s_t)]$$

For the linear value head $V_\theta(s)=w_V^\top\phi_\theta(s)+b_V$:

$$\begin{aligned}\nabla_{w_V}L^V&=\mathbb{E}_t[2(V_\theta(s_t)-V_t^{\text{targ}})\phi_\theta(s_t)]\\ \nabla_{b_V}L^V&=\mathbb{E}_t[2(V_\theta(s_t)-V_t^{\text{targ}})]\\ \nabla_{\theta_\phi}L^V&=\mathbb{E}_t[2(V_\theta(s_t)-V_t^{\text{targ}})w_V^\top\nabla_{\theta_\phi}\phi_\theta(s_t)]\end{aligned}$$

where θ_ϕ are the parameters of the shared feature extractor.

8.1.5 Complete Backpropagation Algorithm

8.2 Computational Graph

8.3 Gradient Clipping and Normalization

Definition 8.5 (Gradient Clipping)**.** To prevent exploding gradients, PPO typically applies gradient clipping:

$$\nabla_\theta L\leftarrow\begin{cases}\nabla_\theta L & \text{if } \|\nabla_\theta L\|\le c_{\max}\\ c_{\max}\frac{\nabla_\theta L}{\|\nabla_\theta L\|} & \text{otherwise}\end{cases}$$

where $c_{\max}$ is typically 0.5 or 1.0.

Definition 8.6 (Advantage Normalization)**.** Within each minibatch, advantages are normalized:

$$\hat{A}_t\leftarrow\frac{\hat{A}_t-\mu_A}{\sigma_A+\epsilon}$$

where μ_A and σ_A are the mean and standard deviation of advantages in the batch, and $\epsilon=10^{-8}$ prevents division by zero.

This normalization:

- Makes the algorithm less sensitive to reward scaling
- Stabilizes training across different environments
- Helps maintain consistent gradient magnitudes

Algorithm 1 PPO Backpropagation

Input: Minibatch $\{(s_t, a_t, \hat{A}_t, V_t^{\text{targ}}, \pi_{\text{old}}(a_t|s_t))\}_{t=1}^{B}$
Initialize: Gradients $\nabla_\theta L = 0$
for $t = 1$ to B **do**
    // Forward pass
    $\phi_t \leftarrow \phi_\theta(s_t)$ ▷ Shared features
    $\pi_{\text{new}}(a_t|s_t) \leftarrow \pi_\theta(a_t|s_t)$ using ϕ_t ▷ Policy output
    $V_t \leftarrow V_\theta(s_t)$ using ϕ_t ▷ Value output
    $r_t \leftarrow \pi_{\text{new}}(a_t|s_t)/\pi_{\text{old}}(a_t|s_t)$ ▷ Importance ratio
    // Compute objectives
    $L_t^{\text{CLIP}} \leftarrow \min(r_t\hat{A}_t, \text{clip}(r_t, 1-\epsilon, 1+\epsilon)\hat{A}_t)$
    $L_t^V \leftarrow (V_t - V_t^{\text{targ}})^2$
    $H_t \leftarrow -\sum_a \pi_\theta(a|s_t) \log \pi_\theta(a|s_t)$ ▷ Entropy
    $L_t \leftarrow L_t^{\text{CLIP}} - c_1 L_t^V + c_2 H_t$
    // Backward pass for policy (actor)
    if $r_t\hat{A}_t < \text{clip}(r_t, 1-\epsilon, 1+\epsilon)\hat{A}_t$ **then**
        $\delta_{\text{policy}} \leftarrow r_t\hat{A}_t \nabla_\theta \log \pi_\theta(a_t|s_t)$
    else
        $\delta_{\text{policy}} \leftarrow 0$ ▷ Clipped, no gradient
    end if
    // Backward pass for value (critic)
    $\delta_V \leftarrow 2(V_t - V_t^{\text{targ}})$
    $\delta_{\text{value}} \leftarrow \delta_V \nabla_\theta V_\theta(s_t)$
    // Backward pass for entropy
    $\delta_{\text{entropy}} \leftarrow c_2 \nabla_\theta H_t$
    // Accumulate gradients
    $\nabla_\theta L \leftarrow \nabla_\theta L + \delta_{\text{policy}} - c_1\delta_{\text{value}} + \delta_{\text{entropy}}$
end for
$\nabla_\theta L \leftarrow \nabla_\theta L / B$ ▷ Average over batch
Return: $\nabla_\theta L$

```
            Input State s_t
                  |
            [Feature Extractor]
             features = phi(s_t)
             /              \
            /                \
   [Policy Head]        [Value Head]
   pi(.|s_t)            V(s_t)
        |                    |
Sample action a_t        Compare with
Compute log pi(a_t|s_t)  V_target
        |                    |
Compute ratio r_t       Compute (V - V_targ)^2
        |                    |
Apply clipping              |
min(r*A, clip(r)*A)         |
        |                    |
        +--------------------+
                  |
             Total Loss
          L = L_CLIP - c1*L_V + c2*H
                  |
            Backpropagate
                  |
      Shared gradients through features
```

Figure 1: PPO computational graph showing forward and backward passes

9 Complete PPO Algorithm

10 Practical Implementation Considerations

10.1 Hyperparameter Selection

10.2 Stability Techniques

1. **Advantage Normalization**: Normalize advantages within each minibatch
2. **Value Function Clipping**: Optionally clip value function updates similar to policy
3. **Observation Normalization**: Maintain running statistics of observations
4. **Reward Scaling**: Normalize rewards or use reward clipping
5. **Learning Rate Annealing**: Decrease learning rate over training

10.3 Common Issues and Solutions

1. **Policy collapse**: Entropy goes to zero, policy becomes deterministic too quickly

Algorithm 2 Proximal Policy Optimization (PPO-Clip)

Input: Initial policy parameters θ_0, initial value function parameters (shared with policy)
Hyperparameters: Clipping ϵ, batch size N, epochs K, minibatch size M, learning rate α, GAE parameters γ, λ
for iteration $= 1, 2, \ldots$ **do**
  // **Data Collection Phase**
  for actor $= 1, \ldots,$ number of parallel actors **do**
    Run policy $\pi_{\theta_{\text{old}}}$ for T timesteps
    Collect trajectory $\{(s_t, a_t, r_t)\}_{t=1}^{T}$
    Compute rewards-to-go: $R_t = \sum_{t'=t}^{T} \gamma^{t'-t} r_{t'}$
    Compute values: $V_t = V_{\theta_{\text{old}}}(s_t)$
  end for
  // **Advantage Estimation Phase**
  for each trajectory **do**
    Compute TD errors: $\delta_t = r_t + \gamma V_{t+1} - V_t$
    Compute GAE advantages: $\hat{A}_t = \sum_{l=0}^{T-t} (\gamma\lambda)^l \delta_{t+l}$
    Compute value targets: $V_t^{\text{targ}} = \hat{A}_t + V_t$
  end for
  Aggregate data: $\mathcal{D} = \{(s_t, a_t, \hat{A}_t, V_t^{\text{targ}}, \pi_{\theta_{\text{old}}}(a_t|s_t))\}$
  // **Policy Optimization Phase**
  for epoch $= 1, \ldots, K$ **do**
    Shuffle $\mathcal{D}$
    for minibatch $\mathcal{B} \subset \mathcal{D}$ of size M **do**
      // Compute importance ratios
      for $(s_t, a_t) \in \mathcal{B}$ **do**
        $r_t \leftarrow \pi_\theta(a_t|s_t)/\pi_{\theta_{\text{old}}}(a_t|s_t)$
      end for
      // Normalize advantages in minibatch
      $\hat{A}_{\mathcal{B}} \leftarrow \text{normalize}(\{\hat{A}_t\}_{t\in\mathcal{B}})$
      // Compute losses
      $L^{\text{CLIP}} \leftarrow \frac{1}{|\mathcal{B}|} \sum_{t\in\mathcal{B}} \min(r_t \hat{A}_t, \text{clip}(r_t, 1-\epsilon, 1+\epsilon)\hat{A}_t)$
      $L^V \leftarrow \frac{1}{|\mathcal{B}|} \sum_{t\in\mathcal{B}} (V_\theta(s_t) - V_t^{\text{targ}})^2$
      $S \leftarrow \frac{1}{|\mathcal{B}|} \sum_{t\in\mathcal{B}} H(\pi_\theta(\cdot|s_t))$
      $L \leftarrow L^{\text{CLIP}} - c_1 L^V + c_2 S$
      // Gradient update
      $g \leftarrow \nabla_\theta L$
      $g \leftarrow \text{clip_gradient}(g, c_{\max})$ ▷ Optional gradient clipping
      $\theta \leftarrow \theta + \alpha \cdot g$ ▷ Gradient ascent
    end for
  end for
  $\theta_{\text{old}} \leftarrow \theta$
  // **Logging and Evaluation**
  if iteration mod eval_interval $= 0$ **then**
    Evaluate policy π_θ and log metrics
  end if
end for

Hyperparameter	Typical Value	Description
Clipping parameter ϵ	0.1 – 0.2	Controls policy update size
Learning rate α	3×10^{-4}	Adam optimizer learning rate
Discount γ	0.99	Reward discount factor
GAE parameter λ	0.95	Bias-variance tradeoff
Epochs per update K	3 – 10	Times to reuse each batch
Minibatch size M	32 – 512	SGD minibatch size
Batch size N	2048 – 4096	Total steps per update
Value loss coef c_1	0.5	Scales value loss
Entropy coef c_2	0.01	Scales entropy bonus
Max grad norm $c_{\max}$	0.5	Gradient clipping threshold

Table 1: Standard PPO hyperparameters

Solution: Increase entropy coefficient c_2, use larger ϵ

2. **Value function divergence**: Value loss increases over time

   Solution: Increase value loss coefficient c_1, use smaller learning rate, implement value clipping

3. **No learning**: Policy doesn't improve

   Solution: Check advantage estimation, verify reward scaling, ensure exploration, increase batch size

4. **High variance**: Training is unstable with large oscillations

   Solution: Increase batch size, decrease learning rate, use more parallel actors, apply gradient clipping

11 Theoretical Guarantees Summary

Theorem 11.1 (PPO Performance Summary)**.** Under standard assumptions, PPO provides:

1. **Approximate Monotonic Improvement** (Theorem 5.3):

$$J(\pi_{\text{new}}) \geq J(\pi_{\text{old}}) - O(\epsilon^2) - O(\delta_A)$$

2. **Convergence to Stationary Point** (Theorem 5.5):

$$\min_{k \leq K} \mathbb{E}[\|\nabla J(\pi_{\theta_k})\|^2] = O(1/\sqrt{K})$$

3. **Sample Complexity** (Theorem 6.2):

$$O\left(\frac{(1-\gamma)^{-3}}{\epsilon^2}\right) \text{ samples to achieve } J(\pi^*) - J(\pi) \leq \epsilon$$

4. **Relative Improvement over Vanilla PG**:

   - $O(1/\epsilon)$ improvement in convergence rate
   - $O(1/(1-\gamma))$ improvement in sample complexity
   - Substantially better empirical stability

12 Conclusion

This document has provided a comprehensive mathematical treatment of Proximal Policy Optimization, including:

- Complete derivation of the clipped surrogate objective from trust region principles
- Rigorous proofs of monotonic improvement and convergence properties
- Detailed sample complexity analysis
- Complete specification of network architecture and backpropagation
- Practical implementation guidance

PPO represents a significant achievement in reinforcement learning: it provides the strong theoretical guarantees of trust region methods while maintaining the simplicity and computational efficiency necessary for practical deep RL applications.

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